

Harriet Fletcher

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MSc Data Science and Artificial Intelligence — Merit | University of Liverpool

Jan 2026

- Thesis: *Stacked ensemble learning for above-ground Carbon Biomass Estimation, Rimba Raya, Indonesia*. Integrated multispectral satellite imagery with field plot ecology data; built a heterogeneous stacked ensemble (RF / SVR / XGBoost + Ridge meta-model); 5-fold nested cross-validation reduced overfitting by ~6%, producing a fully reproducible Python pipeline applicable to large-scale environmental monitoring.

BSc Ecology and Conservation — 2:1 | Lancaster University

July 2023

- Designed and executed independent ecological surveys integrating spatial data collection and statistical analysis (SPSS) across field and laboratory settings.
- Treasurer, Lancaster Environment Centre (LEC) – managed £10,000 annual budget across 12 months ensuring financial compliance, budgeting for five society events.

Projects

Land Acquisition Application for Residential Development | Esri UK Project

April 2026

- Developed a multi-criteria site scoring application in ArcGIS integrating government open data (Housing Delivery Test scores, Land Registry house prices, Environment Agency ecological constraints, OS amenity layers) to rank candidate sites by planning permission viability.
- Built a 3D constraint validation application using Esri tools to cross-reference proposed developments against ecological regulation boundaries, flood risk zones, and amenity proximity thresholds — enabling designers to surface conflicts prior to planning submission.

Regulatory Compliance Checker | Pfizer Pharma Hackathon

March 2026

- Architected a Next.js pipeline that cross-references promotional drug content against regulatory guidelines at build time, surfacing non-compliant copy as flagged warnings and eliminating a class of pre-launch manual review for regulated digital assets.

Distributed Task-Scheduling Architecture | Ultamation / Sciotech Hackathon

Feb 2026

- A 500-task, 5-server automation environment was suffering race conditions and uneven load distribution
- Formalised the problem as a Constraint Satisfaction Problem (CSP) and architected a DAG traversal engine with dependency-aware batching heuristics, eliminating all race conditions and reducing execution latency to the theoretical minimum for the given workload.

Parts of Speech Tagging (POS) | MSc NLP Project

2025

- Engineered an end-to-end clinical NLP pipeline (BiLSTM & Transformer, PyTorch) with torchtext tokenisation and Optuna Bayesian hyperparameter optimisation — Transformer outperformed BiLSTM by 5.8% F1, attributed to self-attention's resolution of long-range dependency loss inherent in sequential hidden state compression.

Geospatial Socioeconomic Analysis | MSc Project

2025

- Applied LISA spatial autocorrelation and DBSCAN clustering to Census API geodata to identify income-inequality hotspots across Austin, TX. isualised in QGIS, demonstrating ArcGIS-equivalent workflows in open-source tooling.

Obesity Classifier for Health Risk Prediction | MSc Project

2025

- Trained a Random Forest classifier on 2,111 clinical records achieving 0.96 F1 / 0.999 AUC; stratified K-Fold CV and correlation-matrix pre-processing addressed class imbalance and multicollinearity.

Mapping Population Change Across Africa | MSc Project

2025

- Reprojected raster data to equal-area CRS for geometrically accurate proportional calculations, applied accessible diverging colour schemes, and implemented tooltip interactivity for spatial exploration of population density change throughout Africa (2015-2025).

Experience

Shared Services Centre Assistant | BDO UK LLP

May – June 2026

- BDO's SSC was migrating multi-client data into CTO2 with duplicate records and unstandardised client name strings creating downstream conflict risks
- Performed systematic verification across large multi-source datasets, identifying structural data quality failures; supported bot-assisted conflict checking incoming clients against the existing portfolio to surface missed conflicts before sign-off.

Accounts Assistant | Bellway plc (FTSE 250)

June – Sept 2025

- Bellway's land acquisition pipeline required rigorous ESG risk assessment across a high-volume development portfolio
- Combined spatial and financial datasets, cross-referencing planning constraints, biodiversity net gain (BNG) metrics, and environmental risk layers at site level; validated BNG datasets for regulated environmental reporting, directly influencing £multi-million strategic land acquisition decisions and increasing fund allocation efficiency by 15%.

Sustainability Analyst | Voiz (Remote)

Oct 2021 – Jan 2022

- Co-developed a Scope 1, 2 & 3 emissions calculator with Cornell University and automated monthly reporting workflows via Excel/Xero integrations
- Reduced reporting turnaround by 5 days and establishing a repeatable, auditable emissions baseline for a company with no prior quantitative measurement capability.

Laboratory Intern | Feedwater Ltd., Wirral

June – Sept 2018

- Engineered an Excel dashboard for microbial treatment review with a dynamic multi-point monitoring interface improving data interpretation speed by 25%, recording 97% biocidal efficacy of bromine in water sterilisation.

Technical stack: Python · PyTorch · JavaScript / TypeScript / Next.js · SQL · QGIS · ArcGIS · GeoPandas · Pandas · Tableau · Power BI · Git

International Teaching (Malaysia, Kerala, Spain, 2023–24) · Full UK Driving Licence